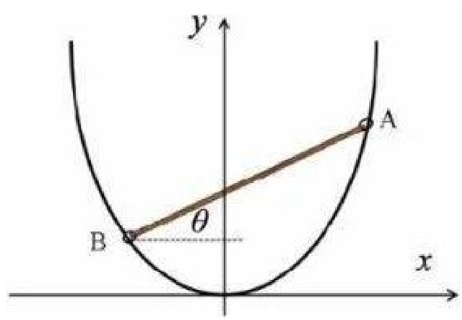
Equilibrium - CPhO 2022

As shown in the figure, a section of a rigid metal wire of a parabolic shape is fixed in a vertical plane. By introducing a horizontal x-axis and a vertical y-axis, the shape of the wire can be described by the equation $y = ax^2$, where a is an unknown constant. A rigid homogeneous rod AB of length $2l$ has small holes at its ends through which a wire is threaded, which allows the rod to slide along it without separation with negligible friction. It turned out that in the equilibrium position the rod is located at an angle $\theta = 30°$ to the horizon. The free fall acceleration is g.



1. Find the constant a.

2. Find the frequency f of small oscillations of the rod about the equilibrium position.

Now let the rod be at rest in the equilibrium position, and a mouse (small enough to ignore its size) begins to climb up. It turned out that all the time the mouse was raised, the rod remained at rest.

1. Find the movement s of the mouse over the rod in time t. Can a mouse climb to the top of the rod? If so, what is the minimum time t_{min} for it to do this?

Collision Challenge : CPhO 2022

Consider the simplest scattering process: particle 1 (incoming particle) with kinetic energy K_1 flies from infinity and collides elastically with particle 2 (target particle) at rest, as a result of which the kinetic energy and direction of motion of the first particle change.

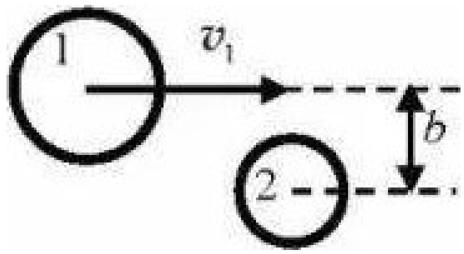
Let the kinetic energy of the particle 1 after scattering be K_1'. We introduce a kinematic factor $k = \frac{K_1'}{K_1}$

1. In what range can the values of k lie?

Let us denote as θ the angle between the directions of particle motion before and after scattering. Let us also introduce the value R — the ratio of the mass of the second particle to the mass of the first. We fix some value of R and look at the behavior of k as a function of θ.

2. Find $k(\theta)$ and specify the range of possible values of θ for an arbitrary fixed R. Consider separately the cases $R > 1, R = 1$ and $R < 1$.

It turns out that in a certain range of values of R, the kinematic factor k can be a multivalued function of θ. To unambiguously determine the branch of the function $k(\theta)$, it is necessary to supplement the description of the interaction of particles. Let's try to do this on the example of the simplest model, in which the colliding particles are considered as uniform smooth spherical solid bodies interacting only upon collision. Let the sum of particle radii be A and the impact parameter be b (see figure).



3. In this model, express k in terms of R, A, and b. Select the branches of this function by considering particular cases of head-on ($b = 0$) and tangent ($b = A$) collisions. Find what range of b values corresponds to each of the obtained branches.

Now let the rest masses of both particles be equal to m_0, the particle sizes are insignificant, and the speed of the incident particle is so high that it becomes necessary to take into account relativistic effects. The speed of light in vacuum is c.

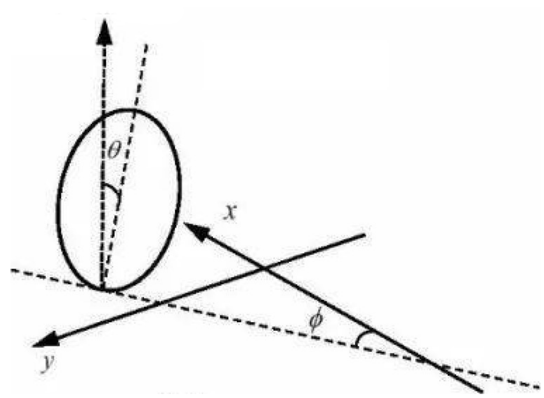
4. Find an expression for the kinematic factor $k(\theta)$. Compare it with the classical result.

5. How are K_1' and the angle α between the directions of particle motion after a collision related? For what K_1' does the function $\alpha(K_1')$ have an extremum? What is the extremum (minimum or maximum) and what is the extreme value of α? What is θ equal to?

Rolling Disk - CPhO 2022

A thin homogeneous rigid wheel with radius R and mass M rolls without slipping along the horizontal plane xy, forming an angle $\theta(t)$ with the vertical. As shown in the figure, the plane of the wheel intersects xy along a straight line forming an angle $\phi(t)$ with the x axis. We denote the coordinates of the point of contact of the wheel with the surface as $(x(t), y(t), 0)$. The free fall acceleration is g.

The functions $x(t), y(t), \theta(t)$ and $\phi(t)$, with which we will describe the movement of the wheel, are not completely independent.



1. Find a kinematic relationship between these functions. The answer can include both the functions themselves and their time derivatives.

Consider a special case of uniform circular motion of the wheel, i.e. when the point of contact with the surface moves at a constant speed, describing a circle of radius r around the vertical axis z. In this case, $\theta(t) = \theta = \text{const.}$, and $\phi(t)$ is a linear function of time. In the laboratory reference frame Σ, without loss of generality, we can say:

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = r \begin{pmatrix} -\sin\omega t \\ \cos\omega t \end{pmatrix},$$

where ω is the angular velocity, which is yet to be found. In the reference frame Σ', which rotates with an angular velocity ω around the z-axis, the angles θ and ϕ are constant, and the wheel rotates around the axis of symmetry that preserves the orientation. In this frame of reference, centrifugal force and the Coriolis force will also act on each section of the wheel.

2. Find the centrifugal force $\vec{F}_\mu$ and Coriolis force $\vec{F}_K$ acting on the wheel, as well as the resulting moments of these forces $\vec{\tau}_\mu$ and $\vec{\tau}_K$, respectively.

3. Find the angular velocity ω and the force $\vec{f}$ acting on the wheel from the side of the surface (ω should not be included in the answer).

If the wheel rolls at a sufficiently high speed, the vertical position $\theta = 0$ may be stable. In this case, the center of mass of the wheel can be considered to move at an almost constant speed V in the positive direction of the x axis. Then, in the first approximation, we can write $x(t) = Vt + \delta x(t)$, where $\delta x(t)$ is a small quantity that changes harmonically with some angular velocity Ω (as do $y(t), \theta(t)$ and $\phi(t)$).

4. Derive in the first non-trivial approximation the equations of motion of the wheel. Find from this the angular velocity Ω and the

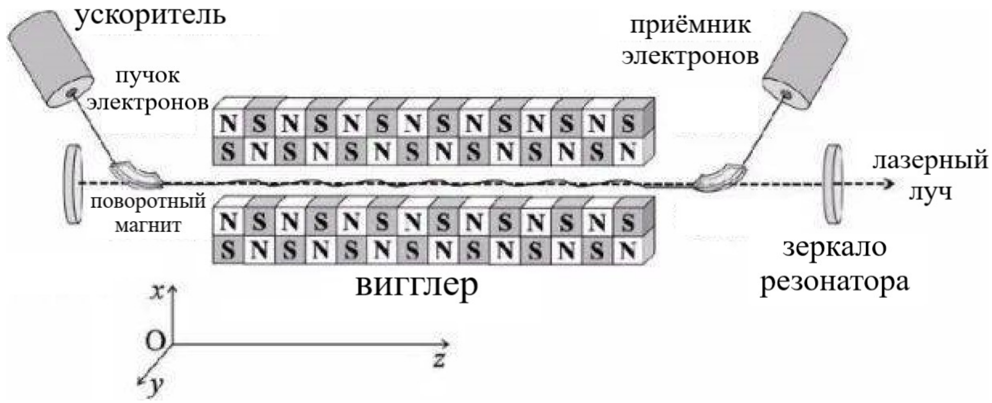
minimum value V_{min} of the wheel speed at which the vertical position will be stable.

Electron Laser - CPhO 2022

A free electron laser is a device that converts the energy of an electron beam into coherent radiation. This technology has great prospects in science, manufacturing and other fields. As shown in the figure below, a free electron laser consists of three main parts: an electron beam accelerator, a wiggler, and an optical cavity. The main part, the wiggler, consists of a series of permanent magnets, the orientation of which changes along the z axis with a spatial period Λ. This allows you to create a constant inhomogeneous magnetic field along the x axis,

$$B = B_0 \cos \frac{2\pi z}{\Lambda}.$$

The electron beam is accelerated in the accelerator to the speed V_0 and directed to the wiggler with the help of a rotating magnet. Under the action of the magnetic field of the wiggler, the electrons begin to make small oscillations along the z axis, losing a small part of their energy to radiation. The rest mass of an electron is $m_e = 9.11 \cdot 10^{-31}$ kg, its charge is $-e$, $e = 1.60 \cdot 10^{-19}$ C. Ignore the effect of gravity.



In addition to the laboratory reference frame $S(x, y, z)$, we introduce a reference frame $S'(x', y', z')$, whose axes are parallel to the S axes and which moves relative to it with the velocity $V_0\vec{e}_z$. At the moment the electron hits the wiggler ($t = t' = 0$), the origins of the coordinates of both systems coincide.
Note: consider the law of transformation of the electromagnetic field as known when switching between reference systems S and S':

$$\begin{cases} E_{x'} = \dfrac{E_x - v_0 B_y}{\sqrt{1 - \left[\frac{v_0}{c}\right]^2}} \\ E_{y'} = \dfrac{E_y + v_0 B_x}{\sqrt{1 - \left[\frac{v_0}{c}\right]^2}} \\ E_{z'} = E_z \end{cases} \qquad \begin{cases} B_{x'} = \dfrac{B_x + \frac{v_0}{c^2} E_y}{\sqrt{1 - \left[\frac{v_0}{c}\right]^2}} \\ B_{y'} = \dfrac{B_y - \frac{v_0}{c^2} E_x}{\sqrt{1 - \left[\frac{v_0}{c}\right]^2}} \\ B_{z'} = B_z \end{cases}$$

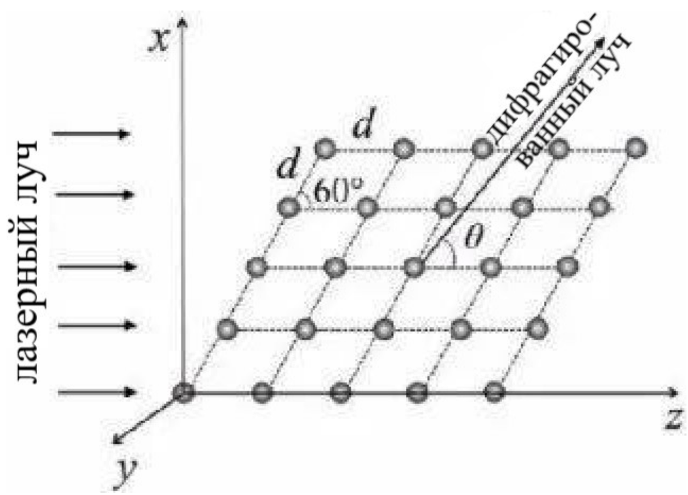
where c is the speed of light.

1. Find in the first approximation the law of electron motion ($x'(t')$, $y'(t'), z'(t')$) in the reference frame S'. Draw qualitatively the trajectory of an electron.

2. Find the accelerating voltage U used to accelerate the electron beam if the spatial period of the wiggler structure is $\Lambda = 1.00$ mm

and the wavelength of the X-ray radiation generated by the laser is $\lambda = 4.00\text{Å}$.

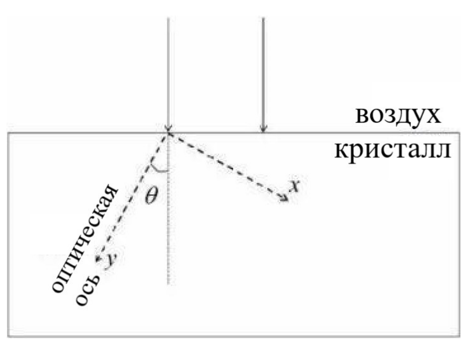
Let now a plane X-ray wave with $\lambda = 4.00\text{Å}$, obtained with the help of this laser, is incident in the xz plane on a rhombic crystal. In this crystal, the distance between neighboring atoms is $d = 8.00\text{Å}$, and the acute angle at the lattice vertices is $60°$.



3. How many main diffraction maxima can be observed at infinity in the xz plane? Find the position of each of them (determined by the angle θ, as in the figure).

Birefringence - CPhO 2022

There are crystals in which the propagation of light occurs anisotropically. In their simplest type - uniaxial crystals - the light is then divided into two waves: ordinary (index o; isotropic) and extraordinary (index e; anisotropic). Also, in uniaxial crystals, there is a distinguished direction - the optical axis of the crystal - during propagation along which the ordinary and extraordinary waves move with the same speed $v_o = \frac{c}{n_o}$, where n_o is the refractive index of the ordinary wave. In this case, the extraordinary wave will move perpendicular to the optical axis with the speed $v_e = \frac{c}{n_e}$. If the direction of propagation of an extraordinary wave is arbitrary, its refractive index turns out to be a continuous function of the angle at which it moves relative to the optical axis.



It follows from the Huygens principle that the front of an extraordinary wave emitted by a point source is an ellipsoid of revolution, the symmetry axis of which is parallel to the optical axis of the crystal. Let's introduce a coordinate system, as shown in the figure, directing the y-axis along the optical axis. Let us consider the propagation of ordinary and extraordinary waves in the xy plane.

1. Guided by the Huygens principle, describe a scheme that allows you to determine the direction of wave propagation in a uniaxial crystal.

Let a plane wave with a wavelength λ normally fall on the surface of a crystal from air. The angle between the normal to the crystal surface and its optical axis is θ.

2. Draw the wavefronts of the ordinary and extraordinary waves propagating in the crystal. Display their wave vectors $\vec{k}_o, \vec{k}_e$ (perpendicular to the fronts) and wave propagation directions $\vec{N}_o, \vec{N}_e$ respectively. For definiteness, consider that $n_o > n_e$.

3. Express the angle ξ between the propagation direction $\vec{N}_e$ of the extraordinary wave and the optical axis of the crystal in terms of n_o, n_e and θ.

The dispersion of the refractive index in a uniaxial BBO crystal (barium β-borate) has the form:

$$\begin{cases} n_o^2(\lambda) = 2.7359 + \dfrac{0.01878}{\lambda^2 - 0.01822} - 0.01354\lambda^2 \\ n_e^2(\lambda) = 2.3753 + \dfrac{0.01224}{\lambda^2 - 0.01667} - 0.01516\lambda^2 \end{cases}$$

(Here, the wavelength λ in vacuum is measured in μm.)

1. Find n_o and n_e in the BBO crystal at vacuum wavelengths of 800.0 nm and 400.0 nm.

One of the key functions of anisotropic crystals is frequency doubling - the transformation of two photons with the original frequency into one photon with doubled frequency. Such a process must satisfy the laws of conservation of energy and momentum. The photon momentum $\vec{p}$ and its wave vector $\vec{k}$ are related by $\vec{p} = \hbar\vec{k}$, where $\hbar$ is the reduced Planck constant.

2. How are the wave vectors of a photon before ($\vec{k}_1$) and after ($\vec{k}_2$) frequency doubling related?

3. Propose a way to satisfy the conservation laws when doubling the frequency in a BBO crystal. The wavelength of the light used in vacuum is 800 nm.

4. At what angle θ would it be possible to double the frequency of this light when passed through a BBO crystal?

5. What will be the angle α between the wave vector $\vec{k}_e$ of the extraordinary wave and the direction of its propagation $\vec{N}_e$?

Ideal $\to$ Real - CPhO 2022

Consider an arbitrary nonrelativistic monatomic gas. If the sizes of gas molecules and their interaction could be neglected, we would come to the model of an ideal gas, but this model is not always applicable to describe real gases. An alternative is the model constructed by van der Waals, in which the equation of state for 1 mole of gas has the form:

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT,$$

where T, P and V are the temperature, pressure and volume of the gas, respectively, R is the universal gas constant, a and b are some positive values. It is easy to see that for any fixed T in the limit $V \to +\infty$ this model goes over to the ideal gas model.

1. What is the physical meaning of the quantity b? Estimate it by considering gas molecules as solid balls of radius r. Avogadro's constant is equal to N_A.

2. Rewrite the equation for N moles of gas. The molar volume of the gas v should not be included in the answer.

It is known that the molar heat capacity of a gas at a constant volume C_V satisfies the condition:

$$\left(\frac{\partial C_V}{\partial v}\right)_T = \left(\frac{\partial^2 P}{\partial T^2}\right)_v,$$

3. Show that $C_V =$ constant and find its value.
   Attention! Do not substitute this result in subsequent calculations!

It is known that the molar internal energy u of the Van der Waals gas is:

$$u = C_V T - \frac{a}{v}.$$

4. Get the expression for the molar entropy $s(T, v)$ of the Van der Waals gas. Write down the equation of the adiabatic process for the Van der Waals gas.
   Note: Your response may contain undefined constants.

Let 1 mole of Van der Waals gas participate in the following Carnot cycle:

- Isothermal expansion at temperature T_1, volume increases from V_1 to V_2;
- Adiabatic cooling - the temperature drops from T_1 to T_2, the volume increases from V_2 to V_3;
- Isothermal contraction at T_2, volume decreases from V_3 to V_4;
- Adiabatic heating - the temperature rises from T_2 to T_1, the volume decreases from V_4 to V_1.

5. Find the heat supplied to the gas (Q_1) and removed from the gas (Q_2). Find the cycle efficiency η and represent it as a function of temperatures T_1 and T_2 only.

Finally, we define the isothermal compressibility factor as:

$$\kappa_T = -\frac{1}{v}\left(\frac{\partial v}{\partial P}\right)_T .$$

6. Get an expression for the isothermal compressibility factor $\kappa_T(T, v)$ of the Van der Waals gas in the limiting case $a << Pv^2$. In what range can the values of κ_T lie? What is the physical meaning of the result obtained?